

**A**  
**Seminar Report**  
**on**  
**HUMAN ORGAN SYSTEMS AND BIO DESIGNS**

**By,**  
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**DEPARTMENT OF ELECTRONICS AND COMMUNICATION  
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**Sapthagiri College of Engineering,**

(Affiliated to Visvesvaraya Technological University and Approved by AICTE)

NAAC Accredited with 'A' Grade

Accredited by NBA

ISO 9001-2015 and 14001-2015 certified Institute

Bengaluru, Karnataka 560057

2025-2026

**A**  
**Seminar Report**  
**on**  
**HUMAN ORGAN SYSTEMS AND BIO DESIGNS**

In partial fulfilment of requirements for the degree of Bachelor of Engineering in Electronics  
& Communication Engineering

**SUBMITTED BY:**

SHARAN N

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**Under the Guidance of**

Prof. PRATHIBHA P



**DEPARTMENT OF ELECTRONICS AND COMMUNICATION  
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**CERTIFICATE**

Certified that seminar work entitled "HUMAN ORGAN SYSTEMS AND BIO-DESIGNS" is a Bonafide work carried out in the Fourth semester by "SHARAN N" in partial fulfilment for the award of Bachelor of Engineering in Electronics & Communication Engineering from Sapthagiri College of Engineering, Bengaluru during the academic year 2025 - 2026.

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Prof. Prathibha P

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4<sup>TH</sup> Semester, ECE

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## CONTENTS

| Sl.No. | Title                      | Pg.No. |
|--------|----------------------------|--------|
| 1.     | Abstract.....              | i      |
| 2.     | Introduction.....          | 01     |
| 3.     | The Brain.....             | 02     |
| 4.     | The Eye.....               | 04     |
| 5.     | The Heart.....             | 06     |
| 6.     | The Lungs.....             | 08     |
| 7.     | The Kidney.....            | 10     |
| 8.     | Seminar Photographies..... | 12     |
| 9.     | Conclusion.....            | 14     |
| 10.    | References.....            | 15     |

## **ABSTRACT**

The study of Human Organ Systems and Bio-Designs plays a crucial role in understanding the intricate functions of the human body and the advancements in biomedical engineering. The human body comprises multiple organ systems, each performing essential functions that sustain life. These systems, including the circulatory, respiratory, digestive, nervous, and musculoskeletal systems, work in coordination to maintain homeostasis.

With advancements in bio-designs, scientists and engineers are developing innovative solutions such as artificial organs, bioprinting, and organ-on-a-chip technology to address medical challenges like organ failure and transplantation shortages. Bio-designs integrate biology with engineering principles, paving the way for regenerative medicine, prosthetics, and tissue engineering. This field not only enhances medical treatments but also pushes the boundaries of biotechnology and personalized medicine.

This seminar explores the structure and function of human organ systems and examines the latest breakthroughs in bio-designs, highlighting their potential to revolutionize healthcare. The integration of technology with biology presents promising solutions to some of the most pressing medical challenges, ensuring a future where science and innovation improve human health and longevity.